

Six Fallacies in Substituting Large Language Models for Human Participants

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Abstract

Can AI systems like large language models (LLMs) replace human participants in behavioral and psychological research? Here I critically evaluate the “replacement” perspective and identify six interpretive fallacies that undermine its validity. These fallacies are: (1) equating token prediction with human intelligence, (2) treating LLMs as the average human, (3) interpreting alignment as explanation, (4) anthropomorphizing AI systems, (5) essentializing identities, and (6) substituting model data for human evidence. Each fallacy represents a potential misunderstanding about what LLMs are and what they can tell us about human cognition. The analysis distinguishes levels of similarity between LLMs and humans, particularly functional equivalence (outputs) versus mechanistic equivalence (processes), while highlighting both technical limitations (addressable through engineering) and conceptual limitations (arising from fundamental differences between statistical and biological intelligence). For each fallacy, specific safeguards are provided to guide responsible research practices. Ultimately, the analysis supports conceptualizing LLMs as pragmatic simulation tools—useful for role-play, rapid hypothesis testing, and computational modeling (provided their outputs are validated against human data)—rather than as replacements for human participants. This framework enables researchers to leverage language models productively while respecting the fundamental differences between machine intelligence and human thought.

Keywords: generative AI (GenAI), large language models (LLMs), AI participants (AI subjects), silicon sampling, simulation or modeling (modelling), research validity

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Collecting data from human participants is resource-intensive and time-consuming. The rise of large language models (LLMs) capable of generating human-like text has sparked substantial interest among psychologists and social scientists (Abdurahman et al., 2025; Demszky et al., 2023; Lin, 2023). In tasks ranging from perception and cognition to language and moral reasoning, LLMs sometimes produce responses that mirror those of average human participants (Aher et al., 2023; Binz & Schulz, 2023a; Dasgupta et al., 2022; Dillion et al., 2023; Hu et al., 2024; Marjeh et al., 2024). As these models are increasingly incorporated into experimental designs and research methodologies, promising instant and tireless responses, a question arises: Could they *ever* stand in for humans in behavioral and psychological research?

Across various fields, proposals have emerged suggesting that LLMs could, to varying degrees, “substitute human participants” in empirical research as “silicon samples” (Sarstedt et al., 2024); “supplant human participants for data collection” in social science as “simulated participants” (Grossmann et al., 2023); serve as “a single participant” and “a proxy for human participants in a certain set of circumstances” in psychological science (Dillion et al., 2023); and “replace humans” in human-centered design to provide “simulated user responses” (Schmidt et al., 2024). Echoing this perspective, commercial entities now peddle synthetic, human-like AI subjects for user and market research—advertising user research “without the users.”

While replacement of human participants is not yet mainstream practice, these emerging viewpoints reflect a broader narrative in AI development—one that envisions systems that could “outperform humans at most economically valuable work” (OpenAI, 2024) or “on a vast array of tasks” (Bengio, 2024). While AI researchers overwhelmingly treat LLMs as task-automation tools and agents, the discourse around such powerful AI capabilities creates a context in which proposals to use LLMs as research participants naturally emerge. In light of these broader technological ambitions and the specific research proposals they inspire, it is both timely and necessary to critically evaluate the replacement perspective and its methodological implications, particularly to clarify the role of LLMs in behavioral and social science research more broadly, whether or not the replacement proposal gains traction.

To this end, this article examines the fundamental assumptions that underlie proposals to use LLMs as human replacements. By clarifying what LLMs actually are—computational tools that simulate linguistic behavior through statistical pattern matching and reinforcement learning (see **Box 1**)—we can better understand their appropriate role in psychology and behavioral sciences.

Six key fallacies are identified, each arising from the misinterpretation of LLMs as human replacements: the token prediction as human intelligence fallacy (conflating statistical text prediction with genuine human intelligence); the average human fallacy (assuming model outputs represent typical human responses); the alignment as explanation fallacy (interpreting similarity between model and human outputs as evidence of shared cognitive mechanisms); the anthropomorphism fallacy (attributing human-like mental states to AI); the identity essentialization fallacy (treating social categories as fixed, homogeneous traits rather than fluid, contextual identities); and the substitution fallacy (presuming LLM-generated data can directly replace human evidence without validation).

These fallacies undermine research validity—the degree to which our evidence and reasoning support the conclusions we draw (Cook & Campbell, 1979; Cronbach & Meehl, 1955; Kerschbaumer et al., 2025; Stanley & Campbell, 1963). Validity concerns in LLM research stem from both methodological choices (how we prompt and test models) and theoretical interpretations (what we claim model outputs represent). Each fallacy undermines

validity in distinct ways: Some compromise our ability to make causal inferences, others limit generalizability across populations, and still others reflect misalignment between what we intend to measure and what we actually capture. The methodological and theoretical safeguards provided for each fallacy address these threats by strengthening both research design and interpretive frameworks.

Taken together, these analyses support the function of LLMs as neural language simulators. Rather than stand-ins for people, they reveal latent psychological and cognitive dynamics embedded in the text—dynamics rooted in the communication of human thoughts, attitudes, and behaviors. This approach offers both conceptual clarity and practical guidance for leveraging LLMs appropriately in behavioral and cognitive research.

Box 1 | Language Models and the Language User Illusion

Modern AI chatbots like ChatGPT are built on large language models (LLMs). Unlike traditional rule-based systems, LLMs process text by converting it into tokens (units of text) and mapping these to numerical vectors in a high-dimensional space (embedding). This mathematical representation enables models to capture semantic relationships between concepts—such as “king” relating to “queen” similarly to how “man” relates to “woman.” The breakthrough transformer architecture (Vaswani et al., 2017) revolutionized these models through attention mechanisms that allow words to gather contextual information from surrounding text. This helps LLMs resolve ambiguities in language and handle complex linguistic tasks that once seemed to require human intelligence.

Base models (also known as “foundation” or “pretrained” models) are typically trained through autoregression—predicting each subsequent token based on preceding ones—enabling them to compress vast amounts of patterns and textual knowledge into their network weights. However, consumer-facing instruct models like ChatGPT undergo further post-training. This often involves supervised learning (fine-tuning on specific instruction–response examples) and reinforcement learning (RL; learning from feedback signals, like rewards or penalties, to improve outputs). One prominent RL technique is reinforcement learning from human feedback (RLHF), which aligns model outputs with human preferences. Beyond alignment, RL methods are increasingly used to enhance specific cognitive-like capabilities, including reasoning and tool use (e.g., interacting with web browsers or code interpreters), often through trial-and-error processes where the models learn to generate more effective or logically sound sequences of thought or action. Beyond model weights, the behavior of instruct models also depends heavily on system prompts (extensive human-authored instructions that guide model behavior), user prompts (Lin, 2024), and inference-time techniques (e.g., methodologies like retrieval-augmented generation [RAG]).

This sophisticated combination contributes to the illusion that the model possesses human-like understanding (Mitchell & Krakauer, 2023). It allows LLMs to model language in ways that go beyond traditional distributional semantics (Lake & Murphy, 2023). They capture nuanced, contextualized meanings and structures in language (Mahowald et al., 2024; Manning et al., 2020; Millièrè, 2024), including visual knowledge embedded in text (Jones et al., 2022; Lewis et al., 2019) and abstract concepts like space and time (Gurnee & Tegmark, 2023). When LLMs produce responses that feel empathetic or insightful, users often project human-like qualities onto them—what can be termed the “language user illusion.” The fluent language generated by LLMs creates a compelling illusion of interacting with another mind, but this overlooks their fundamentally different nature from